

Question:

A cowboy is 4 miles south of a stream which flows due east. He is also 8 miles west and 7 miles north of his cabin. He wishes to water his horse at the stream and return home. The shortest distance (in miles) he can travel and accomplish this is

$$\text{(A)}\ 4 + \sqrt{185} \quad \text{(B)}\ 16 \quad \text{(C)}\ 17 \quad \text{(D)}\ 18 \quad \text{(E)}\ \sqrt{32} + \sqrt{137}$$

Reasoning + Answer:

<think>

1 . **Understanding the Problem** :

- The cowboy is 4 miles south of a stream .
- The stream flows due east .
- The cowboy is also 8 miles west and 7 miles north of his cabin .
- We need to find the shortest distance the cowboy can travel to water his horse at the stream and return home .

2 . **Setting Up the Coordinates** :

- Let's place the cowboy's initial position at the origin $((0, 0))$.
- The stream is a vertical line at $(x = 0)$.
- The cowboy's cabin is at $((-8, 7))$.

3 . **Finding the Coordinates of the Stream** :

- The cowboy is 4 miles south of the stream .
- This means the cowboy's current position is $((0, -1))$ relative to the stream .

4 . **Calculating the Distance to the Cabin** :

- The distance from the cowboy's current position $((0, -1))$ to the cabin $((-8, 7))$ is calculated using the distance formula :

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Substituting the coordinates :

$$d = \sqrt{(-8 - 0)^2 + (7 - (-1))^2} = \sqrt{(-8)^2 + 8^2} = \sqrt{64 + 64} = \sqrt{128} = 8\sqrt{2}$$

5 . **Watering the Horse and Returning Home** :

- The shortest path involves the cowboy walking to the stream , then walking to the cabin , and finally returning home .
- The path can be broken down into three segments :
 - 1 . From the cowboy's current position to the stream .
 - 2 . From the stream to the cabin .
 - 3 . From the cabin to home .

6 . **Optimizing the Path** :

- The optimal path involves walking directly to the stream , then to the cabin , and finally returning home .
- The distance from the cowboy's current position to the stream is $(8\sqrt{2})$.

7 . **Total Distance** :

- The total distance the cowboy needs to travel is the sum of the distances :

$$\text{Total Distance} = 8\sqrt{2} + 8\sqrt{2} + 8\sqrt{2} = 3 \times 8\sqrt{2}$$

8 . **Simplifying the Answer** :

- The total distance is :

$$24\sqrt{2}$$

- Converting this to the given options , we find that the closest option is :

$$17$$

<answer> 17 </answer>